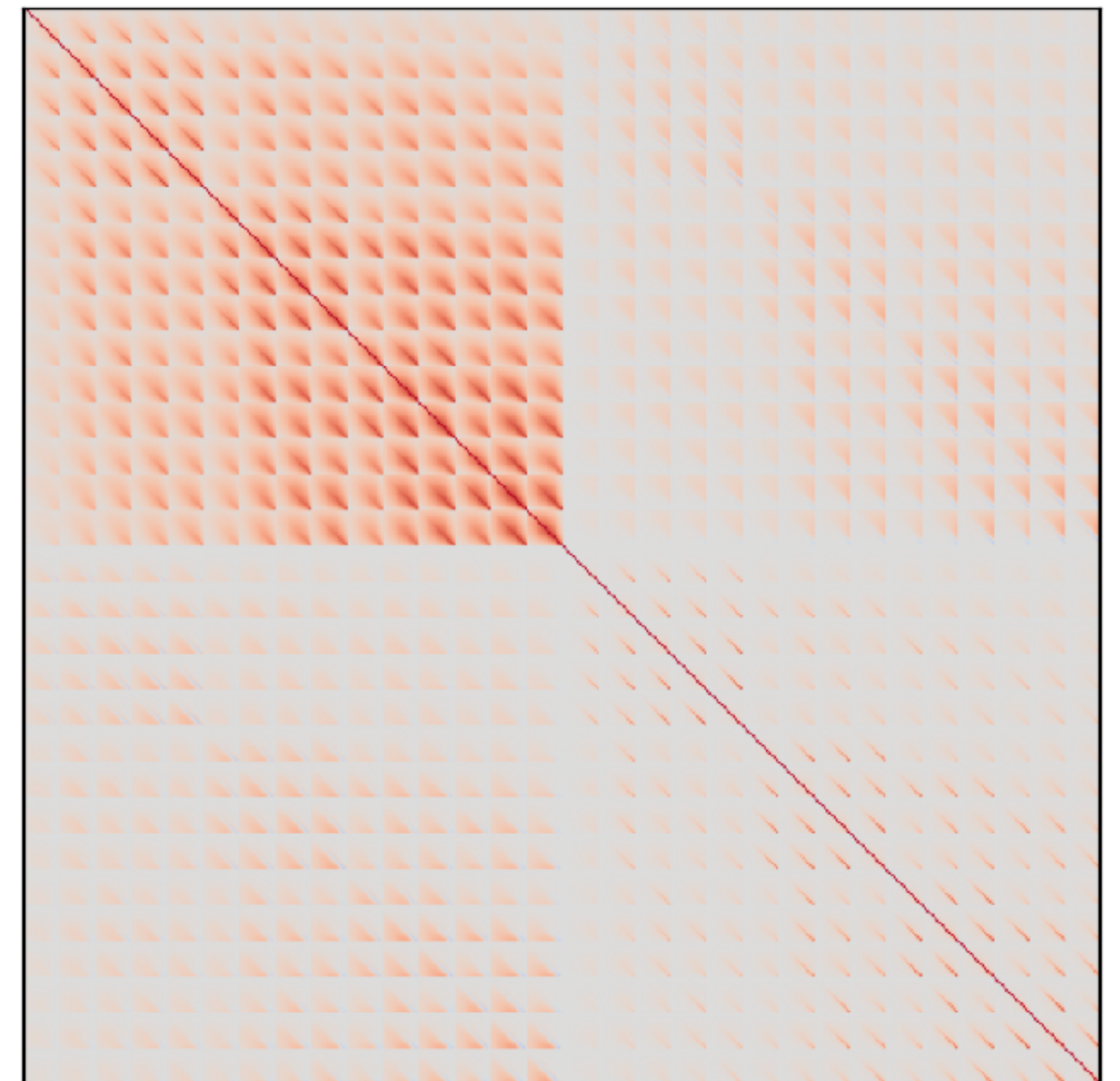


Covariance Matrix Estimation

Jed Homer LMU | ODSL



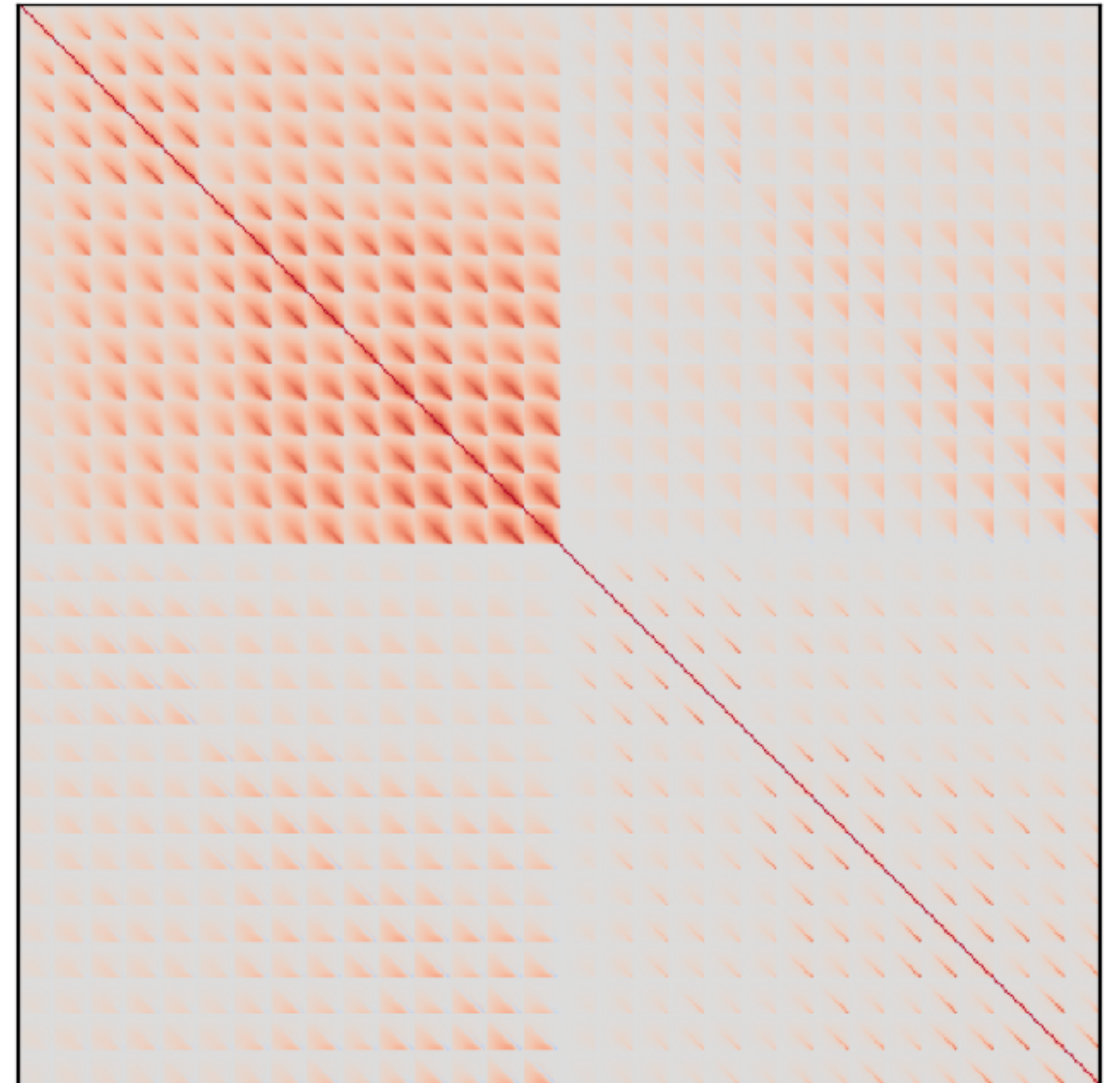
Σ



Synopsis

- What is a covariance matrix, and why is it so important?
- What happens to posteriors if we don't know it?
- If we can accurately simulate it, how do we account for noise in the estimate?

Σ



Likelihood PDF

- Inference requires a likelihood **PDF** which sets the limit of **information in $\hat{x}$ on π**

$$p(\hat{x} \mid x[\pi], \pi)$$

- Data $\hat{x}$ scatters around the expectation $x[\pi]$ as a function of π

Likelihood PDF

- Inference requires a likelihood **PDF** which sets the limit of **information in $\hat{x}$ on π**

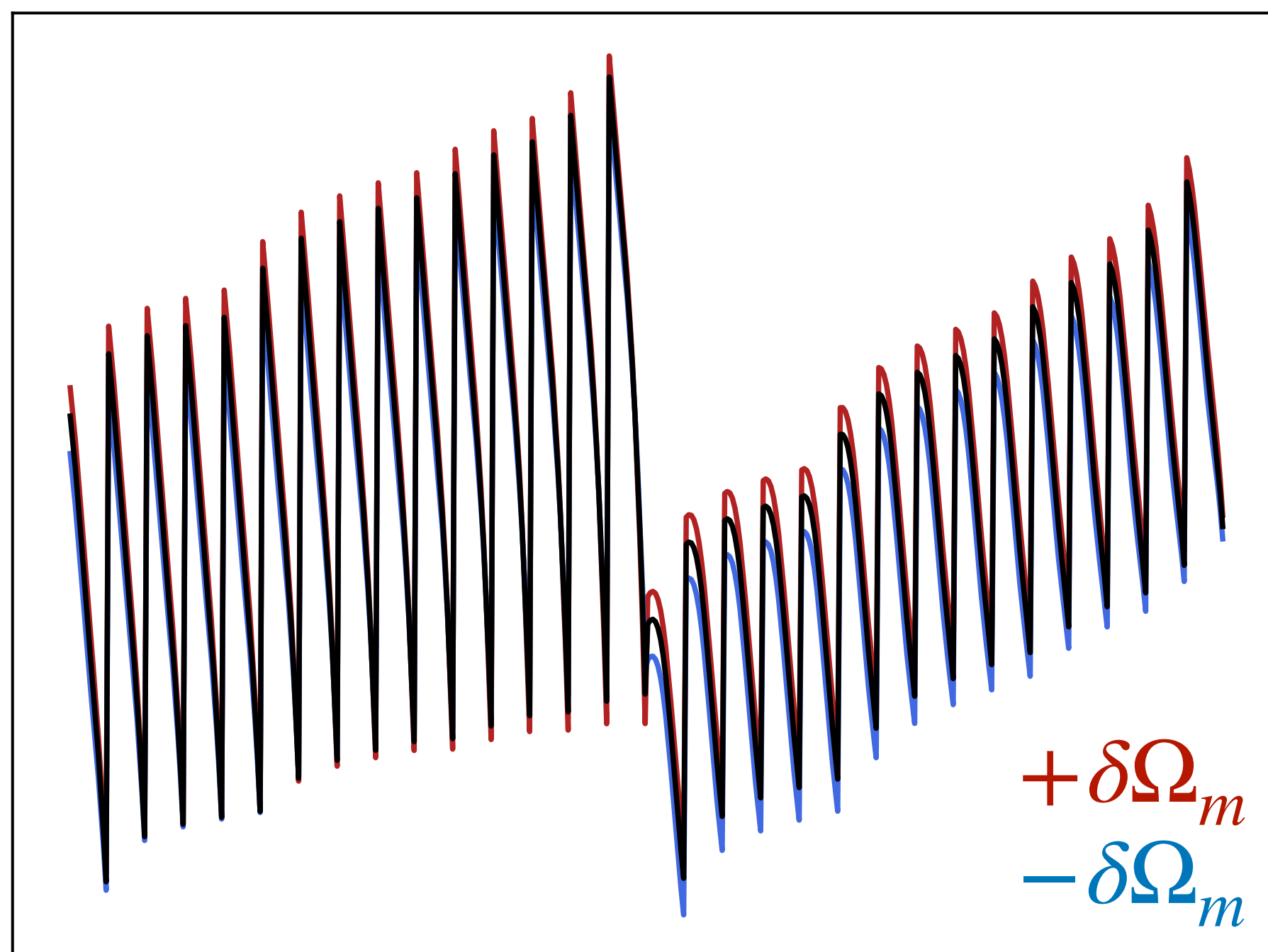
$$p(\hat{x} \mid x[\pi], \pi)$$

e.g. $\mathcal{G}[\hat{x} \mid x[\pi], \Sigma]$

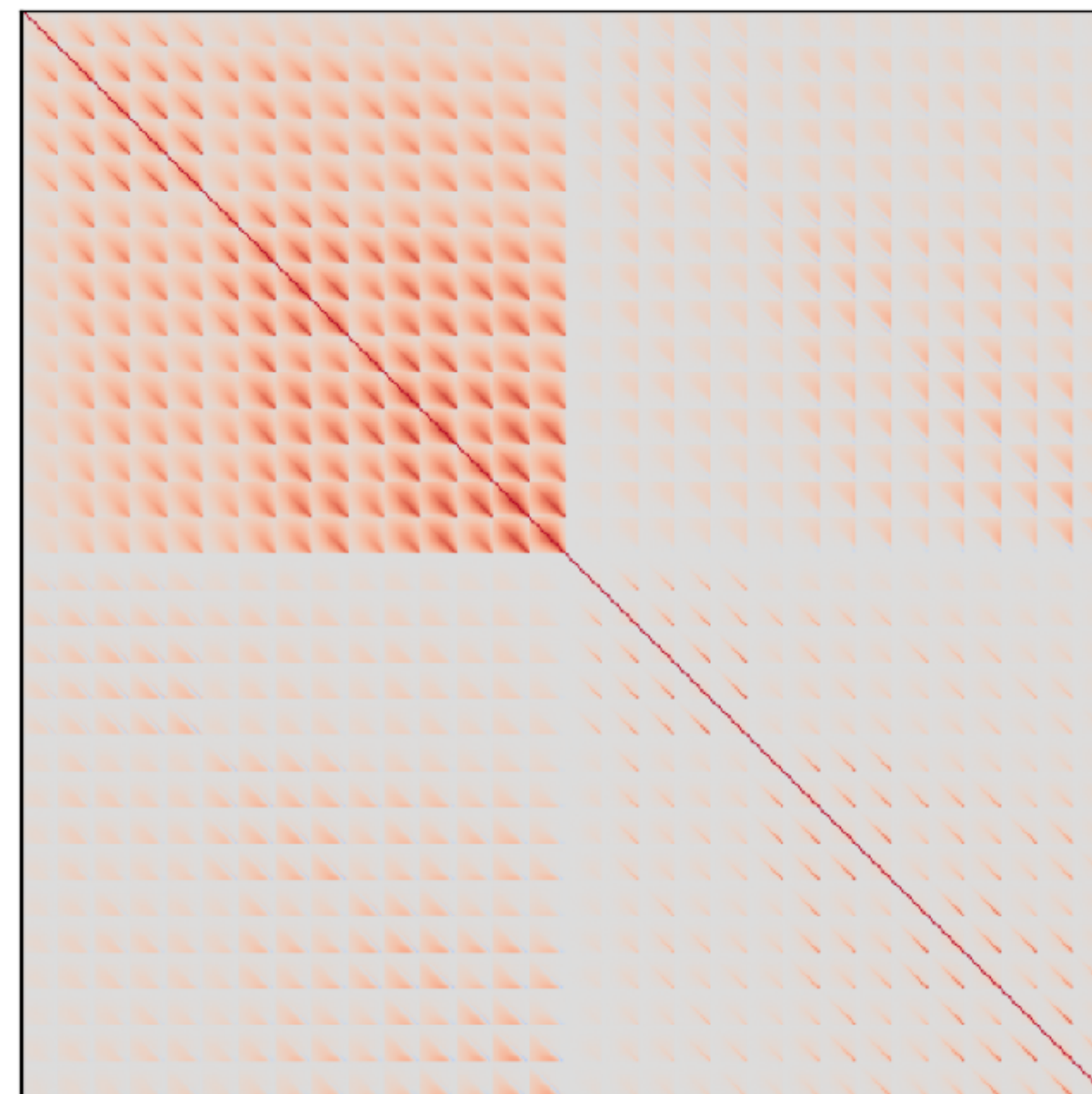
$$\propto \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

$$p(\hat{x} \mid \pi) = \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$

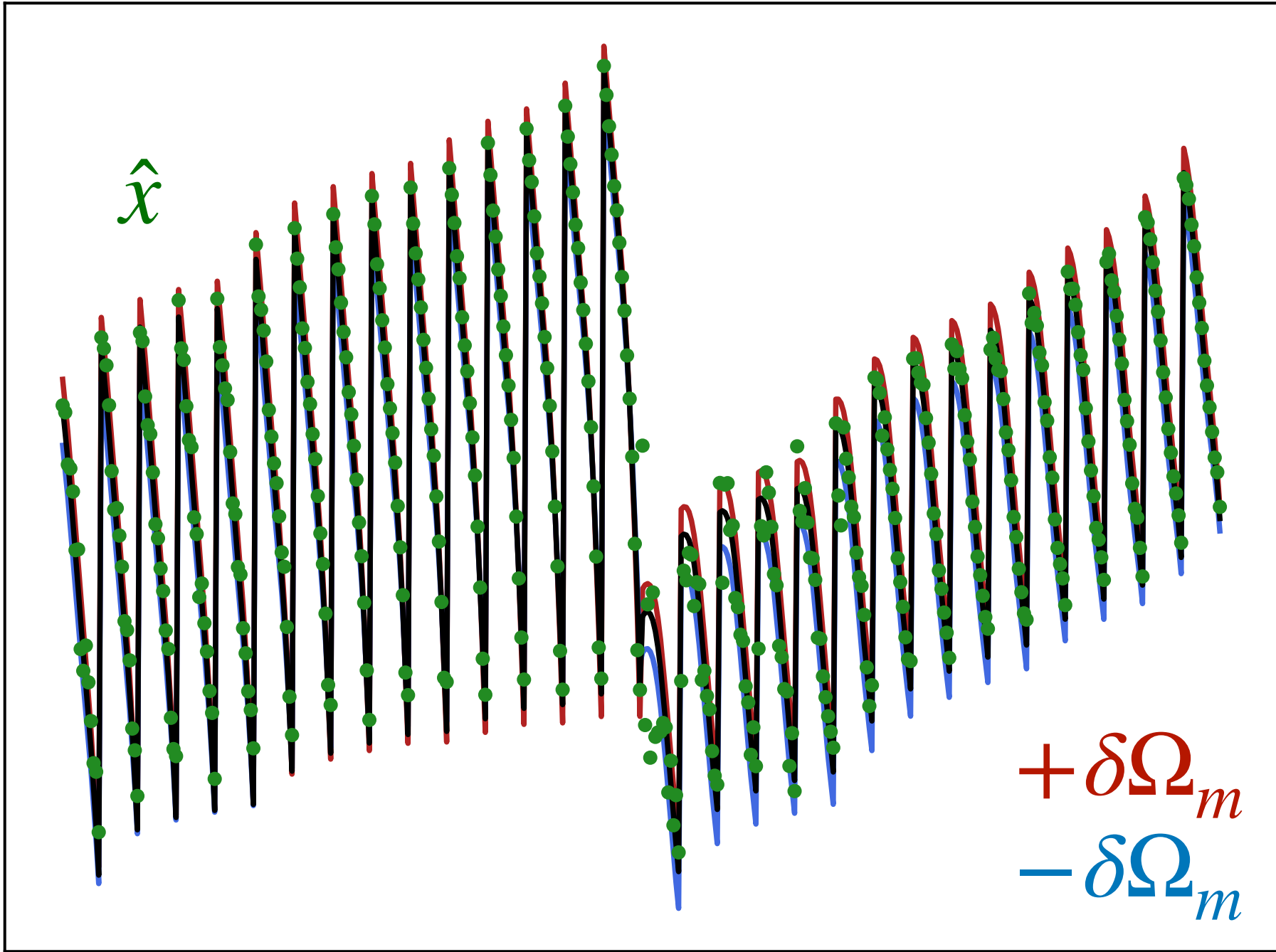


Σ

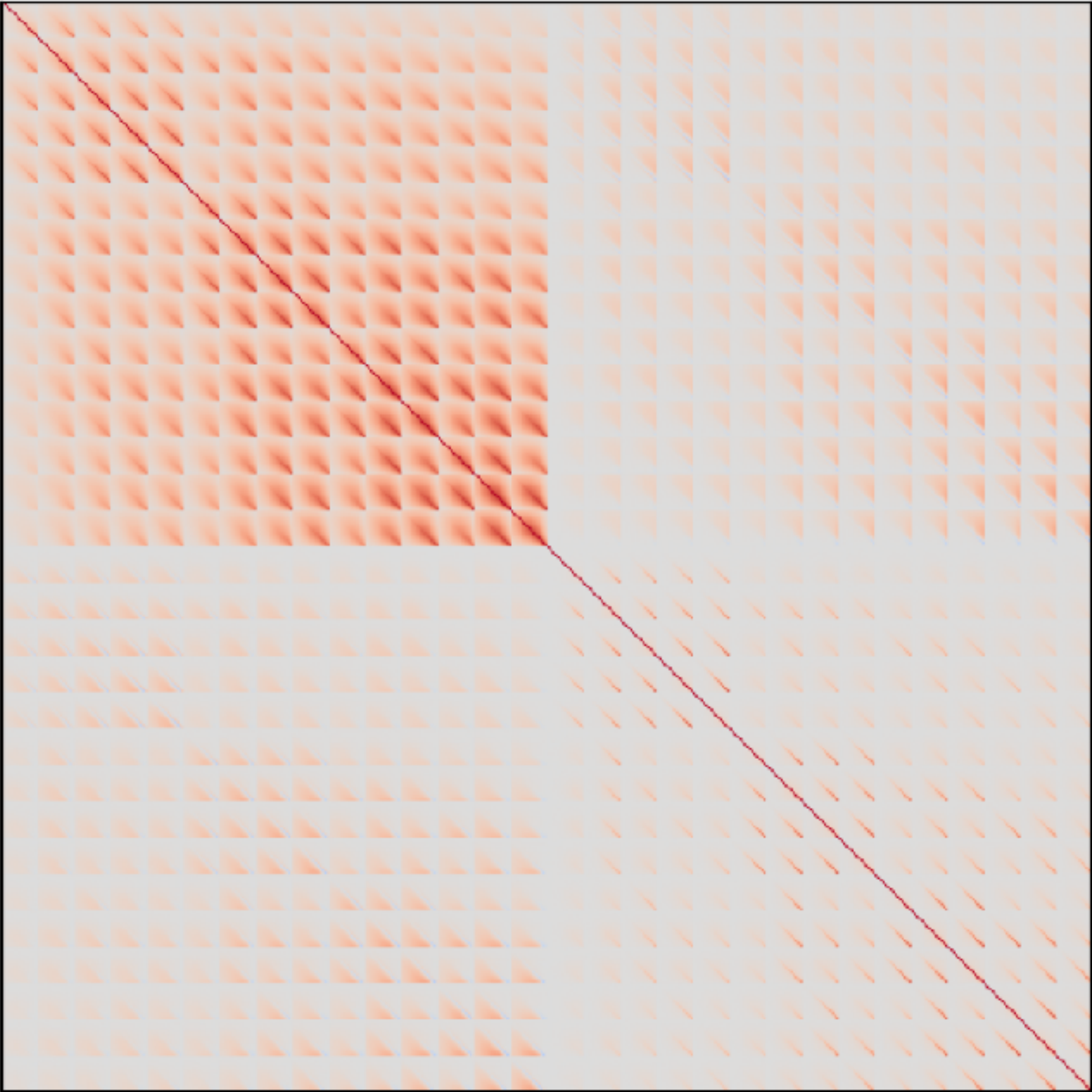


$$\hat{x} \sim \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$



Σ



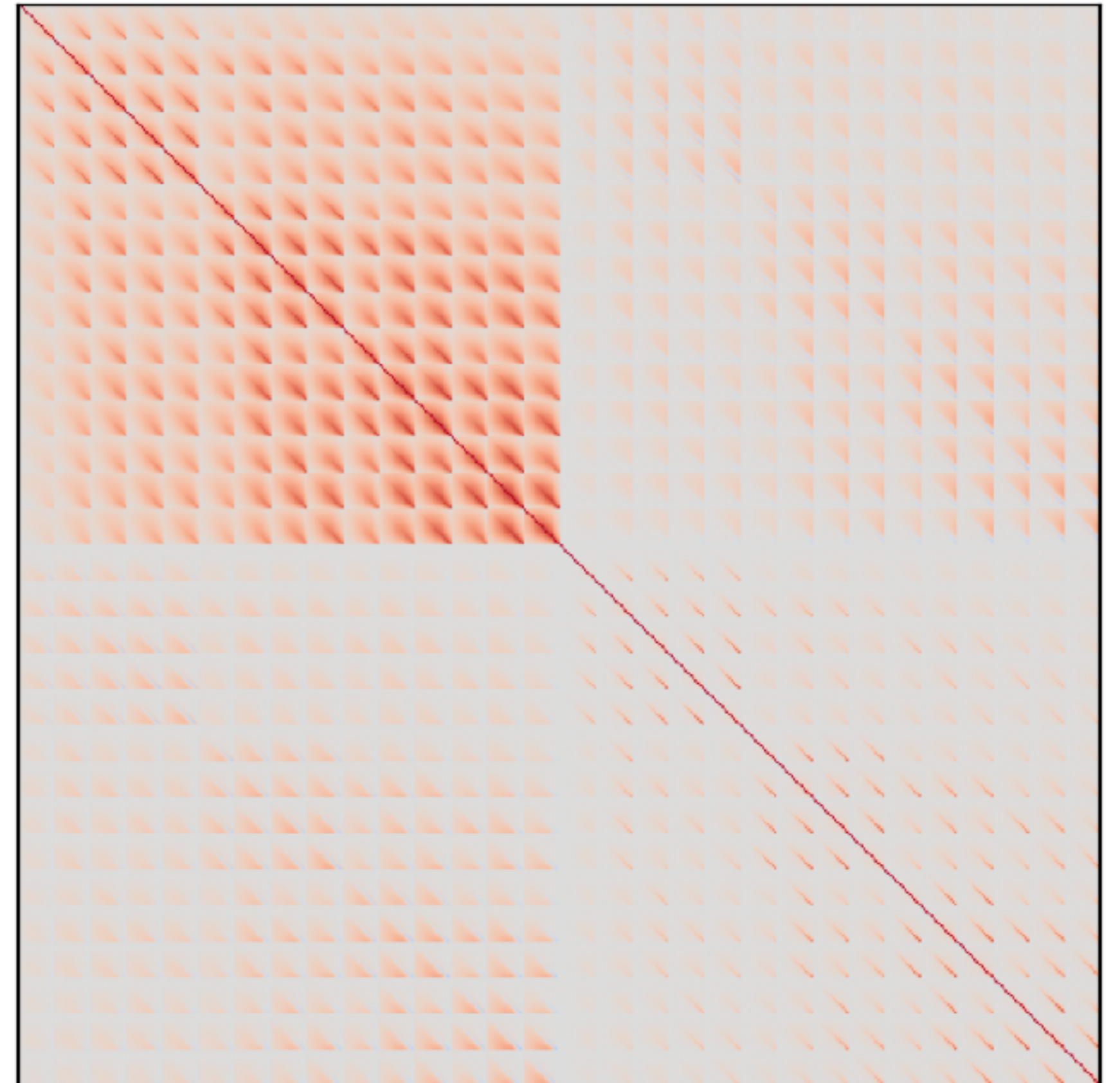
Covariance matrix Σ

- MLE estimator

$$\hat{\Sigma} = \frac{1}{n_s - 1} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$$

- Sampling distribution $p(\hat{\Sigma} | \Sigma) = \mathcal{W}[\hat{\Sigma} | \Sigma, \nu]$

Σ



Where do we obtain $\hat{\Sigma} \approx \Sigma$ from?

- Nature doesn't give us Σ , there are 3 ways to obtain an estimator $\hat{\Sigma}$

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Generate simulations of the data, their scatter estimates the covariance.

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2. Internal covariance estimation

Subsets of the data probe the scatter in the data, the covariance.

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1. Covariance estimation from simulated data

Generate simulations of the data, their scatter estimates the covariance.

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Subsets of the data probe the scatter in the data, the covariance.

3. Theoretical covariance model

Derive a theory-based model for the covariance from first principles.

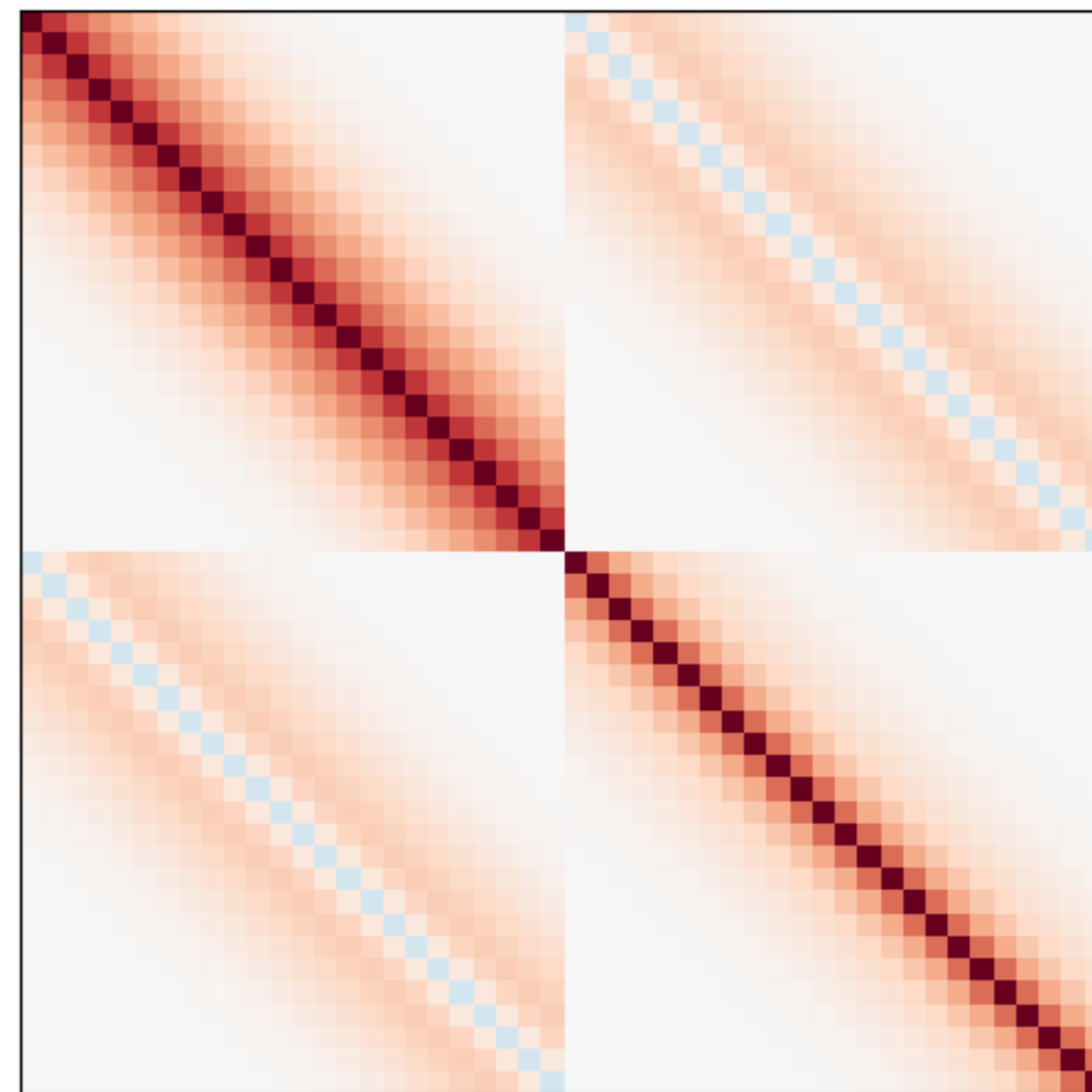
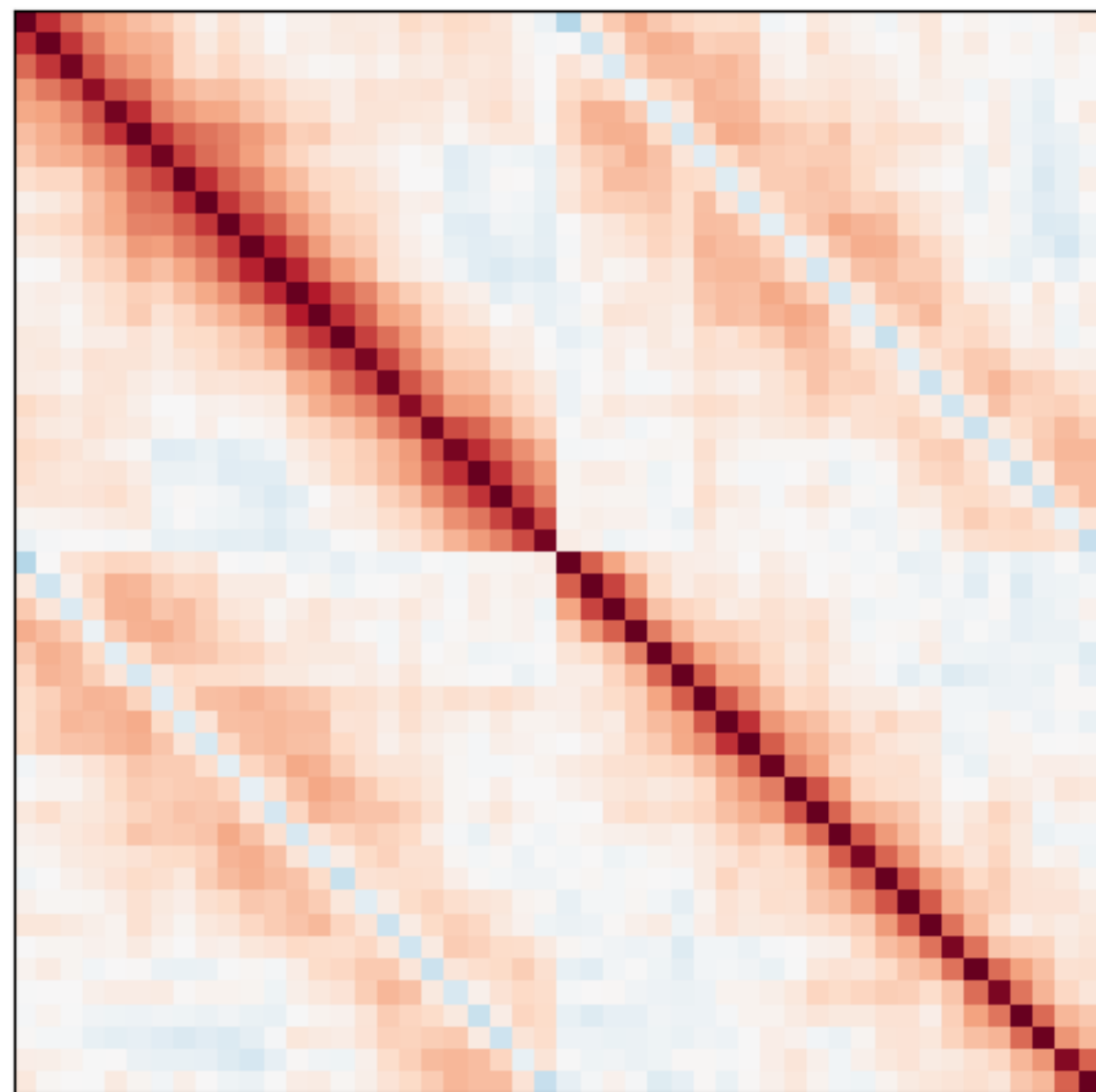
Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

- Let's say we have a set of n_s simulations $\mathcal{X} = \{x_i, \dots, x_{n_s}\} \sim \mathcal{G}[x | x[\pi], \Sigma]$.
- We then use the MLE $\hat{\Sigma} = \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$.
- What happens when we invert this noisy matrix $\hat{\Sigma}$?

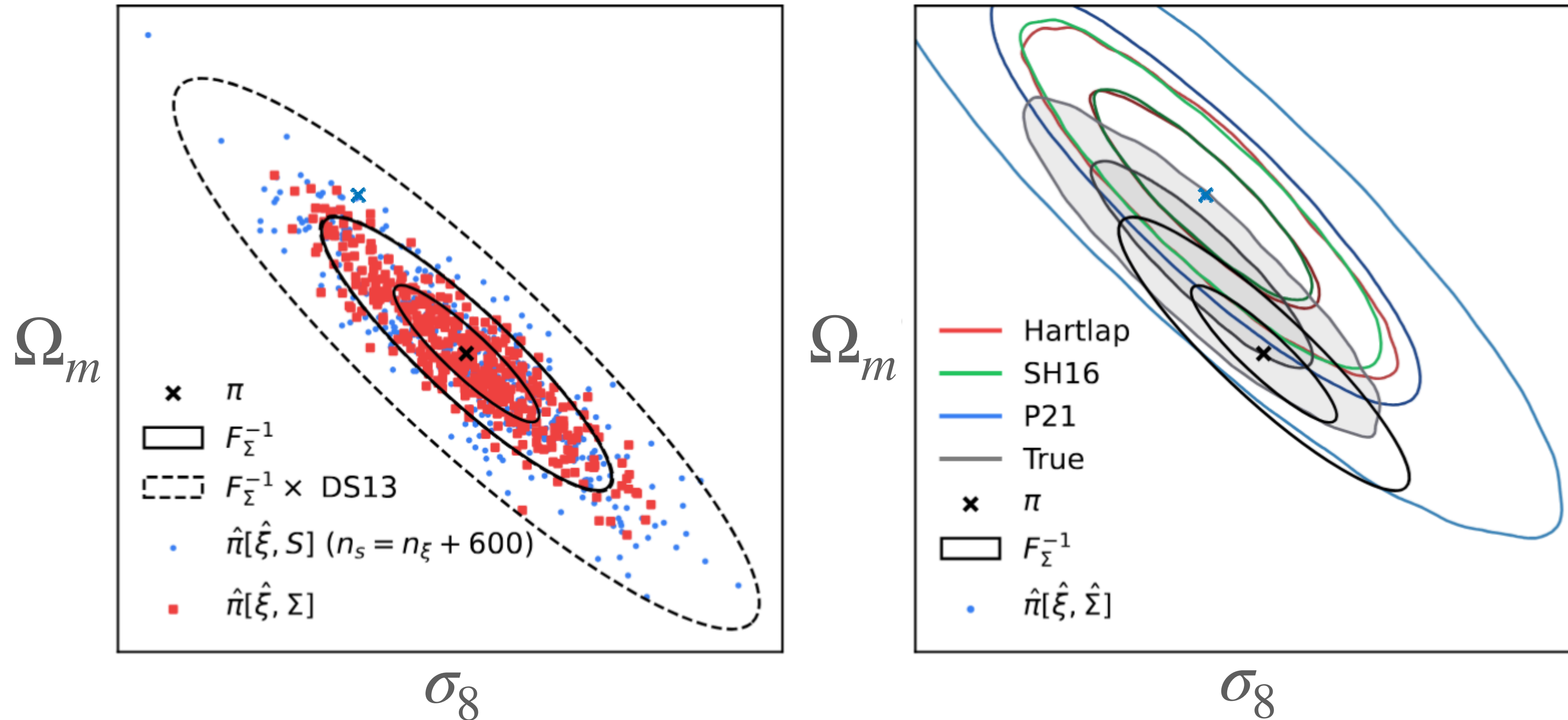
Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

$$\hat{\Sigma} = \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$$

$n_s = 285$



Problem: estimating π from $\hat{x}$ when $\hat{\Sigma} \neq \Sigma$



Noise in $\hat{\Sigma} \neq \Sigma$ adds scatter to best-fit $\hat{\pi}[\hat{x}, \hat{\Sigma}]$. Not accounted for in Gaussian ansatz.

Distribution of $\hat{\Sigma}$

- If our data x are Gaussian, the distribution of $\hat{\Sigma}$ is given by a Wishart distribution

$$p(\hat{\Sigma}) \propto |\hat{\Sigma}\Sigma^{-1}|^{\frac{n_s - n_x - 2}{2}} \exp\left[-\frac{n_s}{2}\text{Tr}(\hat{\Sigma}\Sigma^{-1})\right]$$

Expectation of $(\hat{\Sigma})^{-1}$

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- The expectation of the $\hat{\Psi} = (\hat{\Sigma})^{-1}$ is a multiple of the true precision

$$\mathbb{E}[\hat{\Psi}] = \frac{n_s - 1}{n_s - n_x - 2} \Sigma^{-1}$$

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This shows us how to unbiased an estimated data covariance! Known as the Kaufman-Hartlap factor.

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Covariance of the covariance

- If our data x are Gaussian, the distribution of $\hat{\Sigma}$ is given by a Wishart distribution

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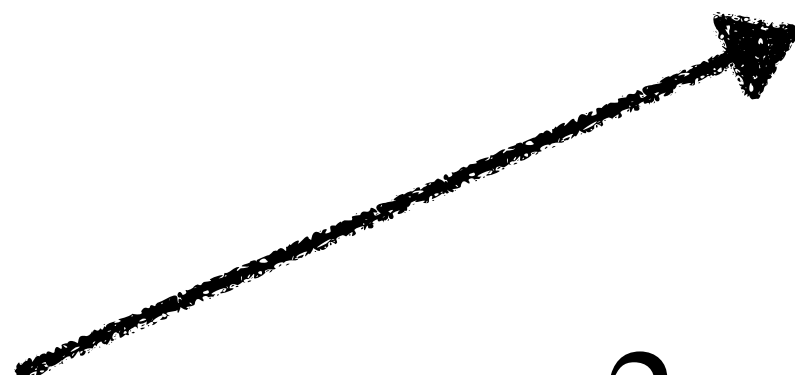
- The covariance of the elements in the estimated covariance is given by

$$\text{Cov}[\hat{\Sigma}_{ij}, \hat{\Sigma}_{kl}] = \frac{1}{n_s - 1} (\Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk})$$


The covariance of off-diagonal elements still depends on the diagonal elements!

Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

- What happens when we invert this noisy matrix $\hat{\Sigma}$?
- $\hat{\Psi}$ depends on the determinant of $\hat{\Sigma}$
 - this is a non-linear combination of $\hat{\Sigma}$ elements
 - this strongly amplifies the noise in the inverse covariance estimate $\hat{\Psi}$...
- This **biases** the estimator $\hat{\Psi}$!

$$\text{Var}[\hat{\Psi}_{ii}] = \frac{2}{n_s - n_x - 4} \Psi_{ii}^2$$


$p(x \mid \pi, \hat{\Psi})$ is a non-linear function of $\hat{\Psi}$

- We are interested in the likelihood $p(x \mid \pi)$, but we're approximating it with the use of $\hat{\Psi}$
- Our likelihood is biased, because $p(x \mid \pi)$ is non-linear in $\hat{\Psi}$

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \mathbb{E}[\hat{\Psi}])]$$

- This means our likelihood is incorrect...

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \Sigma^{-1})]$$

$p(x \mid \pi, \hat{\Psi})$ is a non-linear function of $\hat{\Psi}$

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$$\mathbb{E} [p(x \mid \pi,$$

- This means our likelihood is

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \Sigma^{-1})]$$

We can estimate this bias by the *Percival Correction* (applied to our posterior covariance)

$$P(n_s, n_x, n_\pi) \approx 1 + \frac{n_\pi + 1}{n_s - n_x}$$

Maximum-likelihood estimators $\hat{\pi}$

- Let $\mu = x[\pi]$ so we have $p(x | \pi) = p(x | x[\pi], \pi)$ in general.
- Deriving this equation, showing its covariance...

$$\hat{\pi} = \pi + M[\hat{\Psi}] (\hat{x} - x[\pi])$$

Cov $[\hat{\pi}]$

- Let's derive the covariance of this MLE
- It equals the Fisher information at π , $F[\pi]$. What does this mean?

$$\hat{\pi} = \pi + M[\hat{\Psi}] (\hat{x} - x[\pi])$$

Best-fit $\hat{\pi}$ with estimated covariance $\hat{\Sigma} \neq \Sigma$

- Estimating **best-fit** $\hat{\pi}$ from $\hat{x}$ with an **estimated covariance** $\hat{\Sigma}$

$$\hat{x} \sim \mathcal{G}[\hat{x} | x[\pi], \Sigma]$$



$$\hat{\pi} = \pi + M[\hat{\Psi}] (\hat{x} - x[\pi])$$

$\hat{\Psi}$

Σ^{-1}

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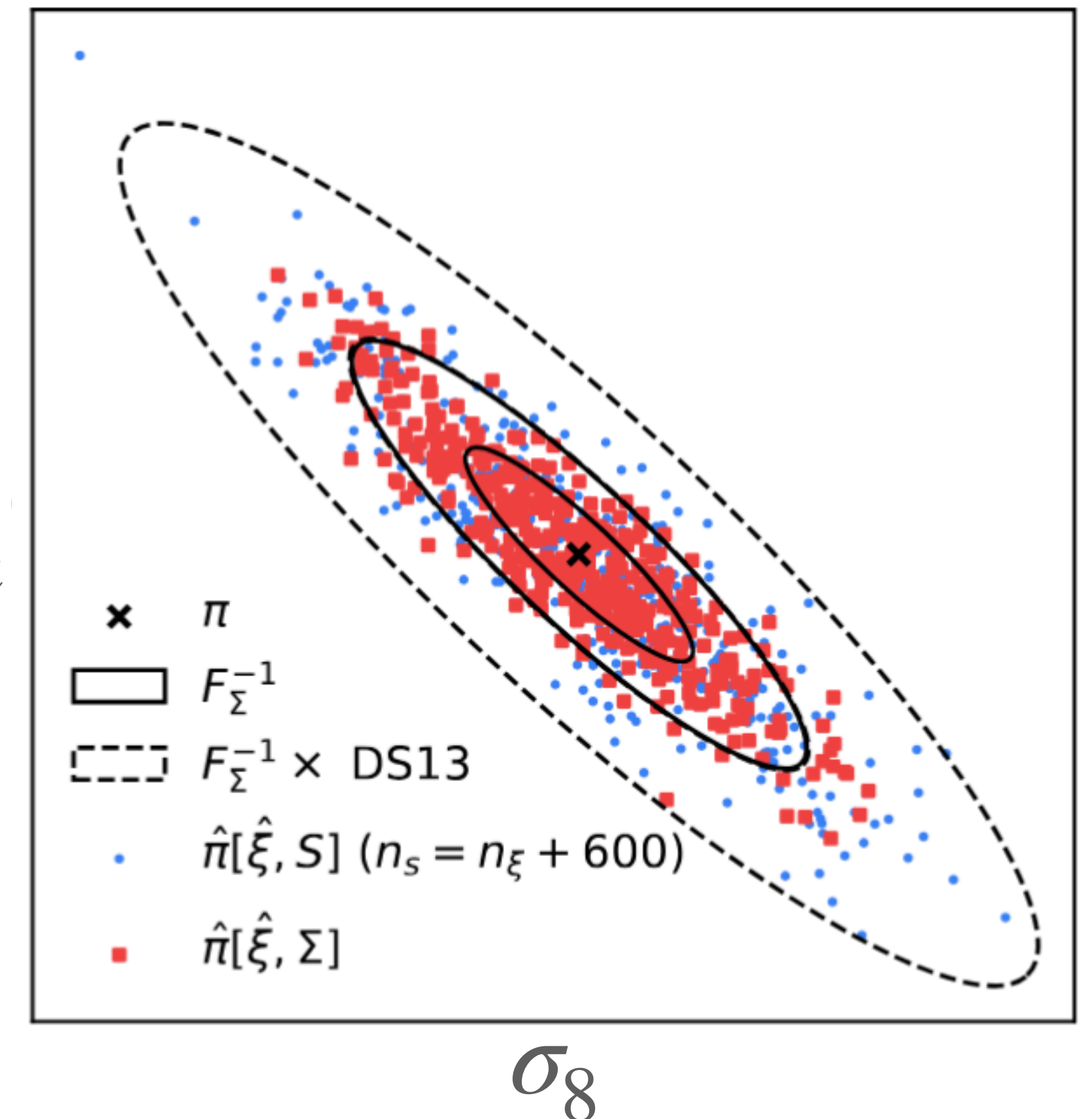


$$\hat{\pi} = \pi + M[\hat{\Psi}] (\hat{x} - x[\pi])$$

$\hat{\Psi}$

Σ^{-1}

Ω_m

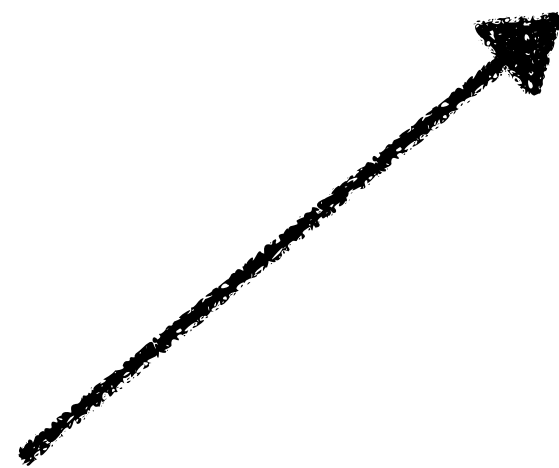


The Dodelson-Schneider Effect

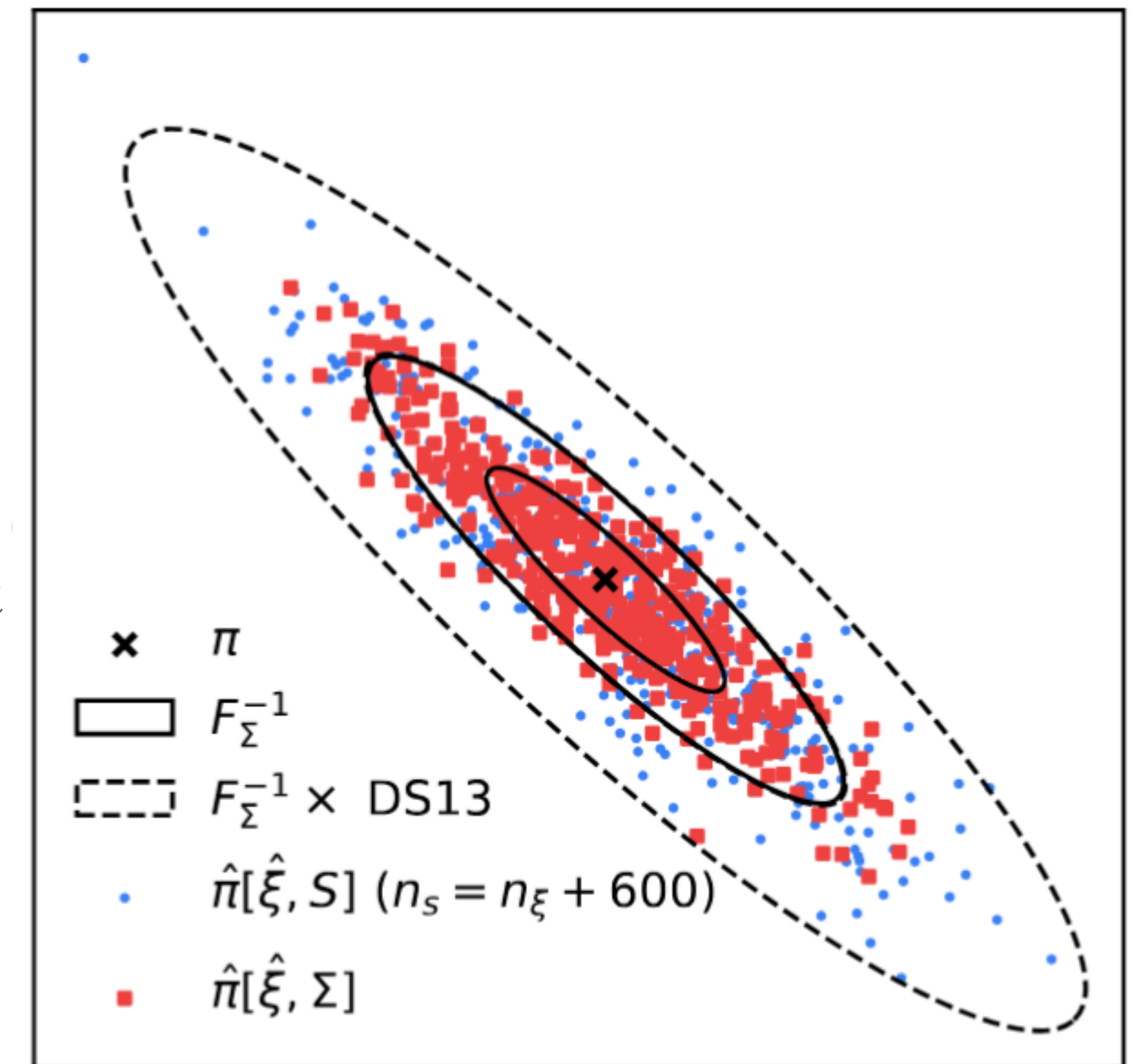
- Estimate the covariance $\hat{\Sigma}$ with n_s simulations

$$\hat{\pi} = \pi + M[\hat{\Psi}] (\hat{x} - x[\pi])$$

$$\hat{\Psi} = \Sigma^{-1} + \Delta_{\Sigma}$$



Ω_m



σ_8

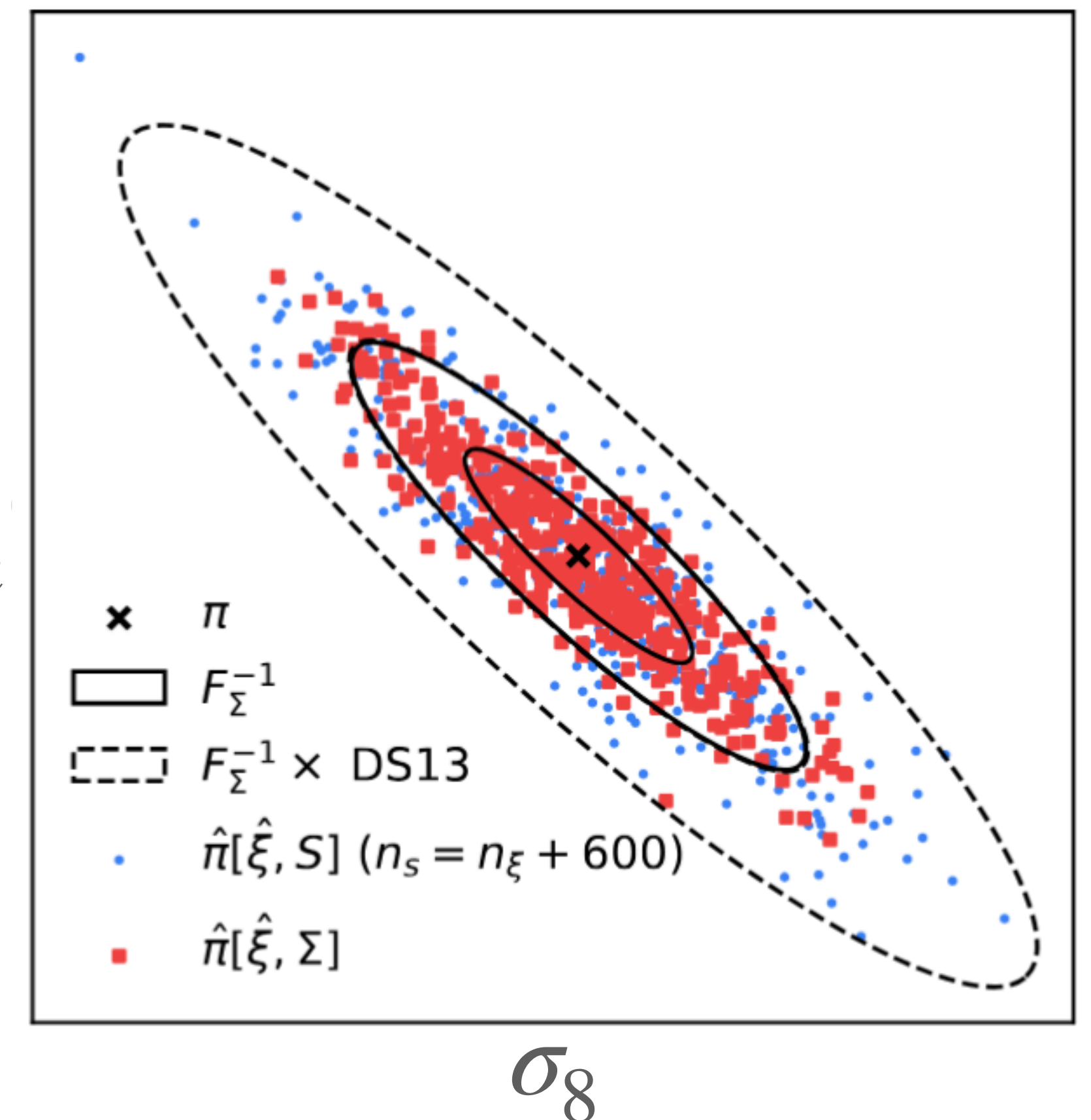
The Dodelson-Schneider Effect

- Estimate the covariance $\hat{\Sigma}$ with n_s simulations

$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}](\hat{x} - x[\pi])$$

$$\hat{\Sigma}_{\pi} = \Sigma_{\pi} \left[1 + \frac{n_x - n_{\pi}}{n_s - n_x} \right]$$

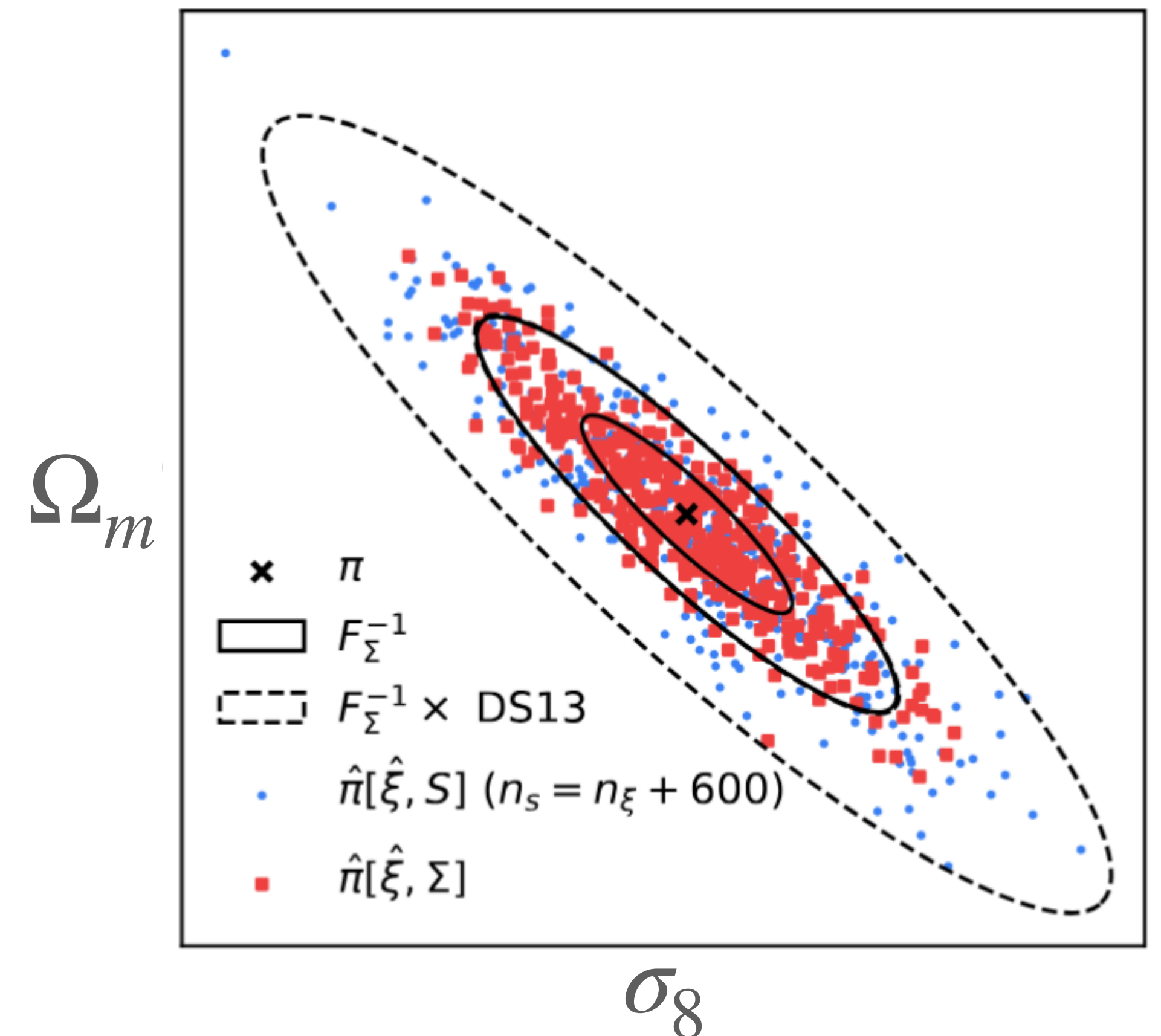
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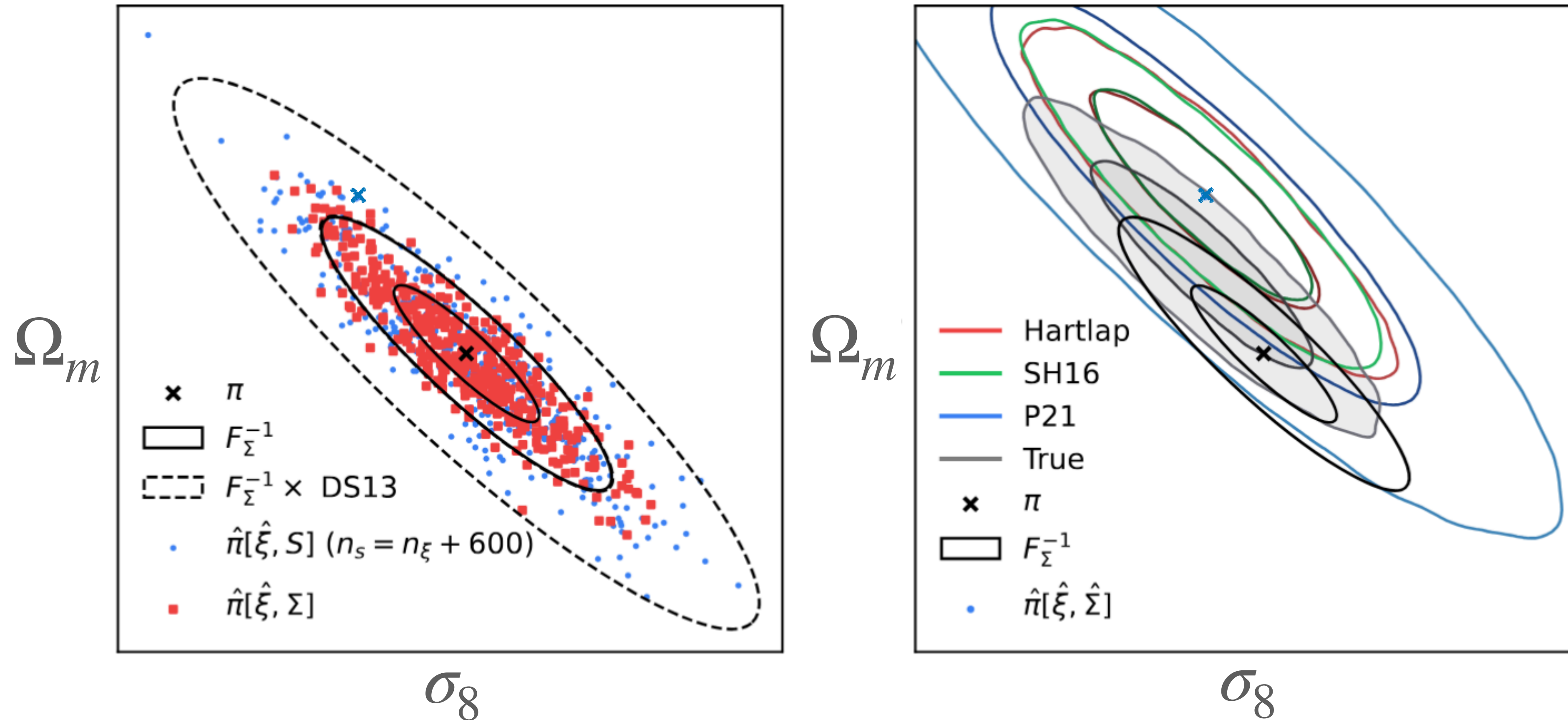
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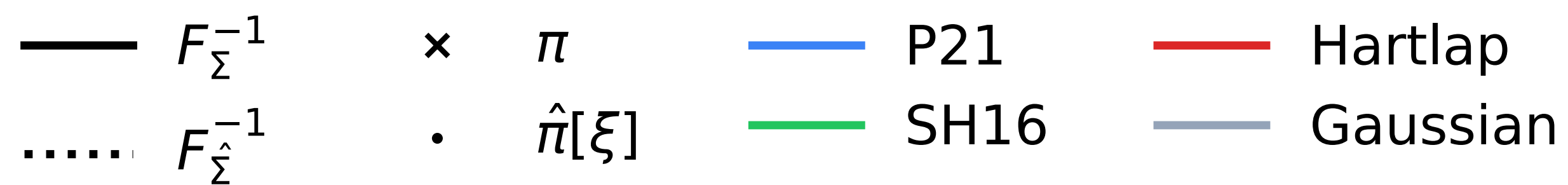
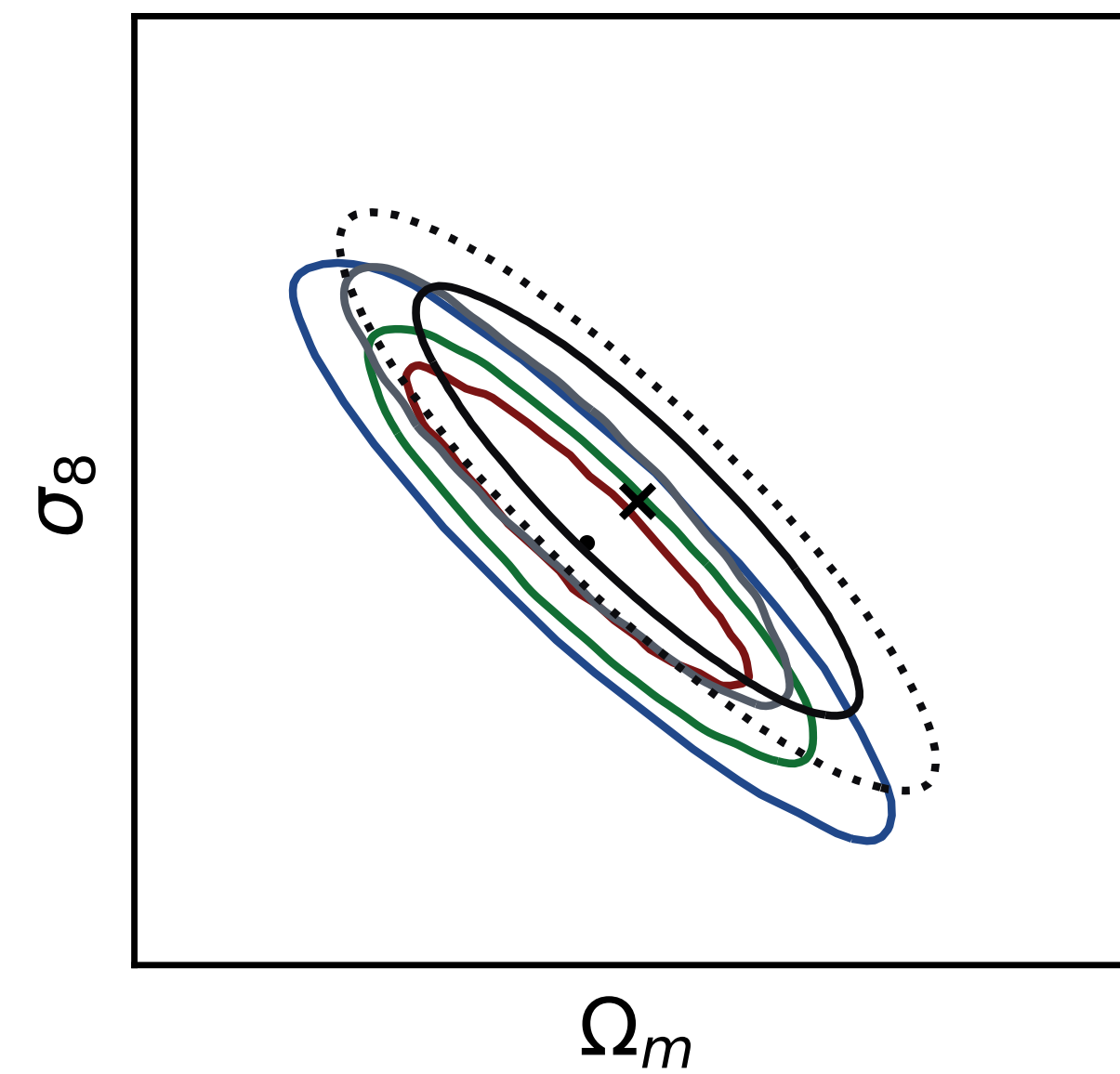
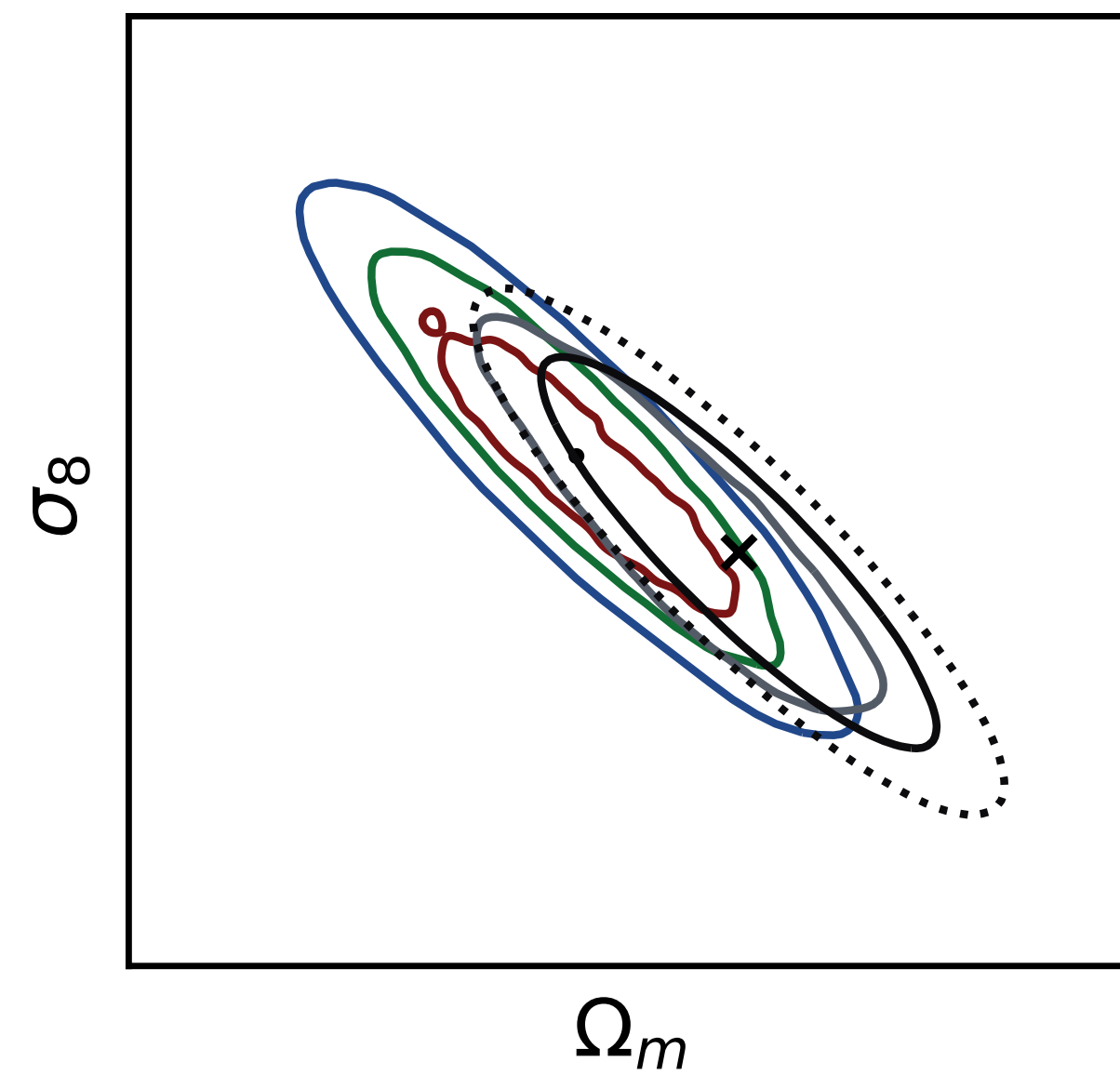
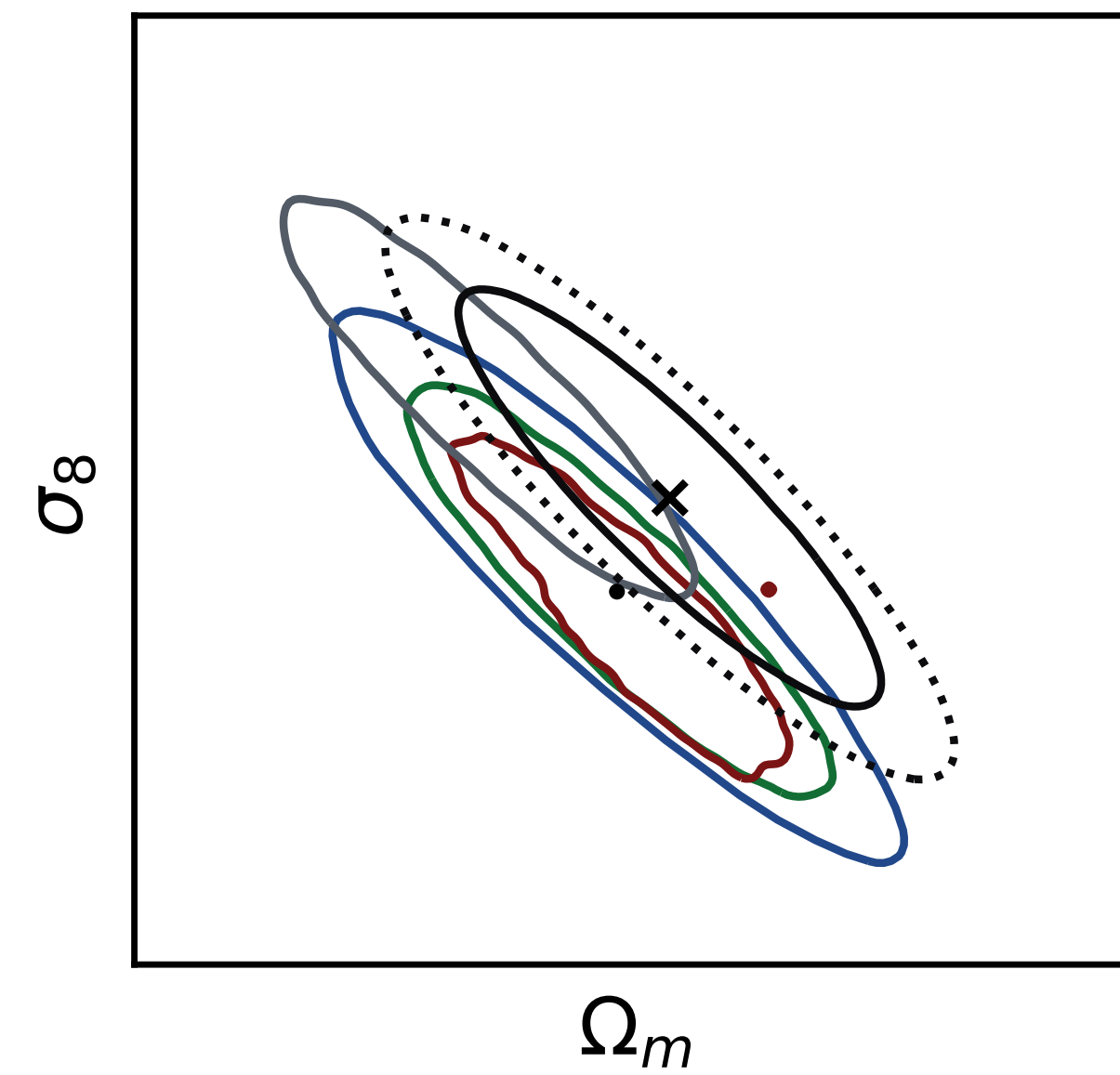
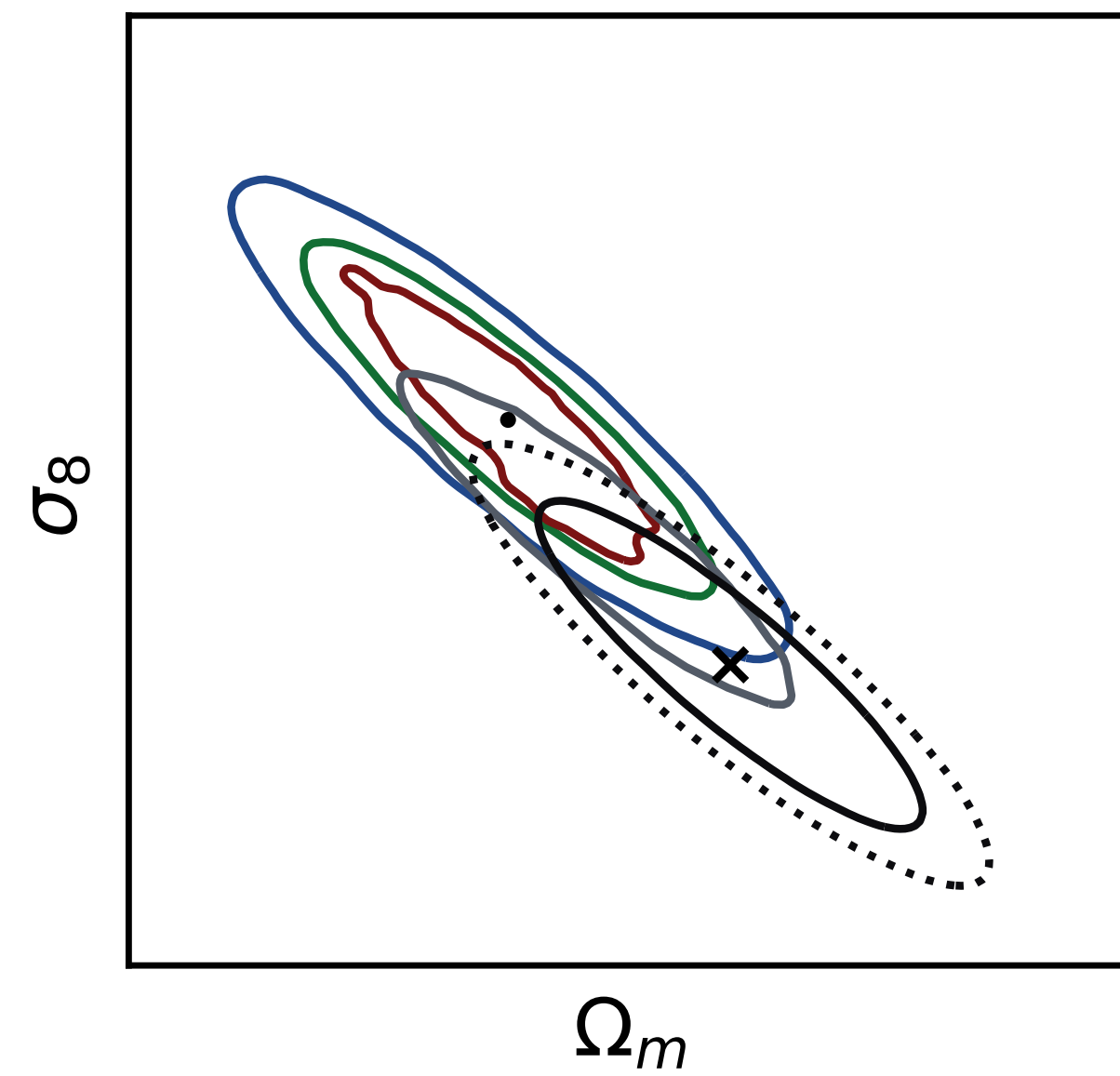
- What does this mean?
- Limits?
- Can we fix it?



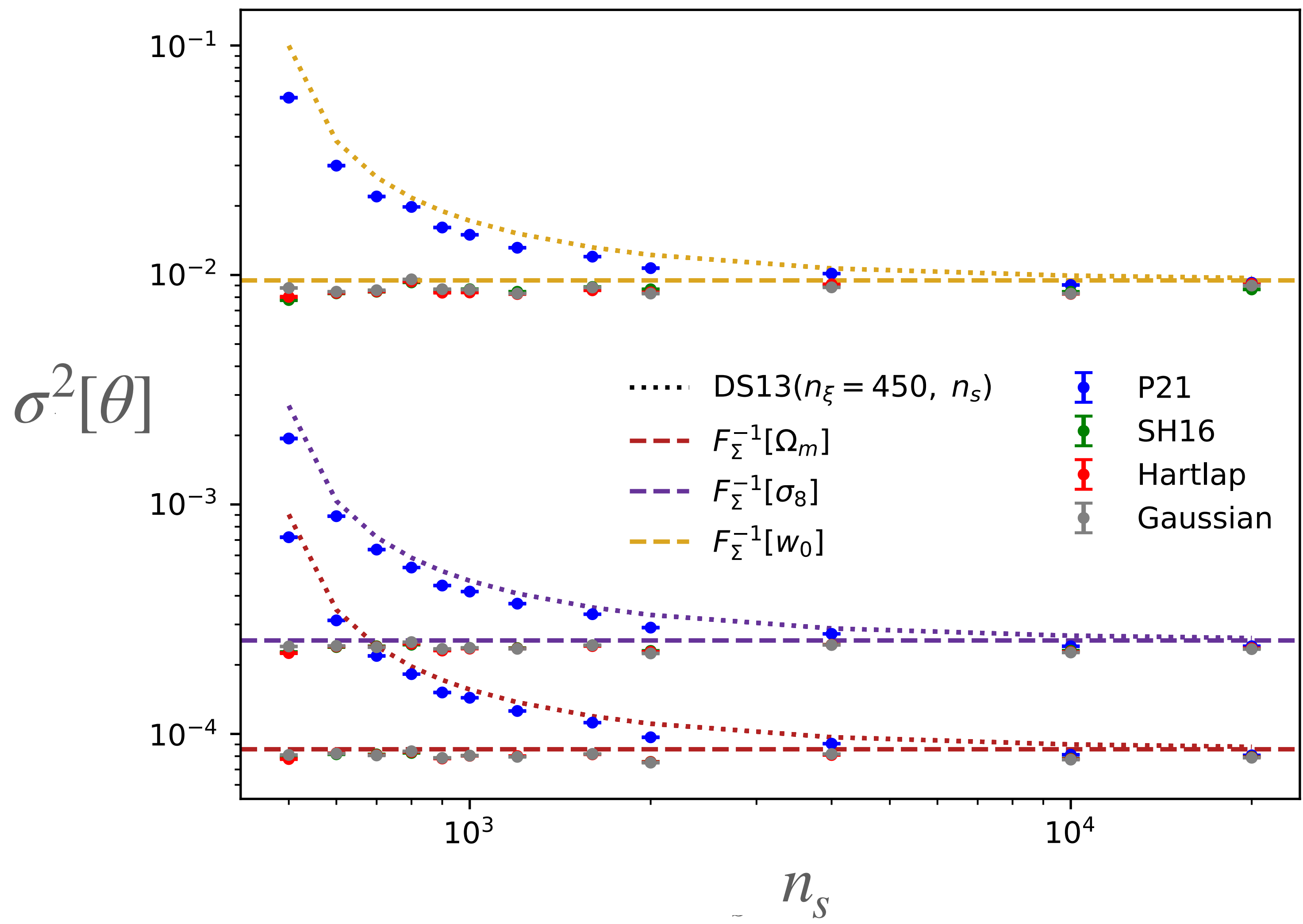
Problem: estimating π from $\hat{x}$ when $\hat{\Sigma} \neq \Sigma$



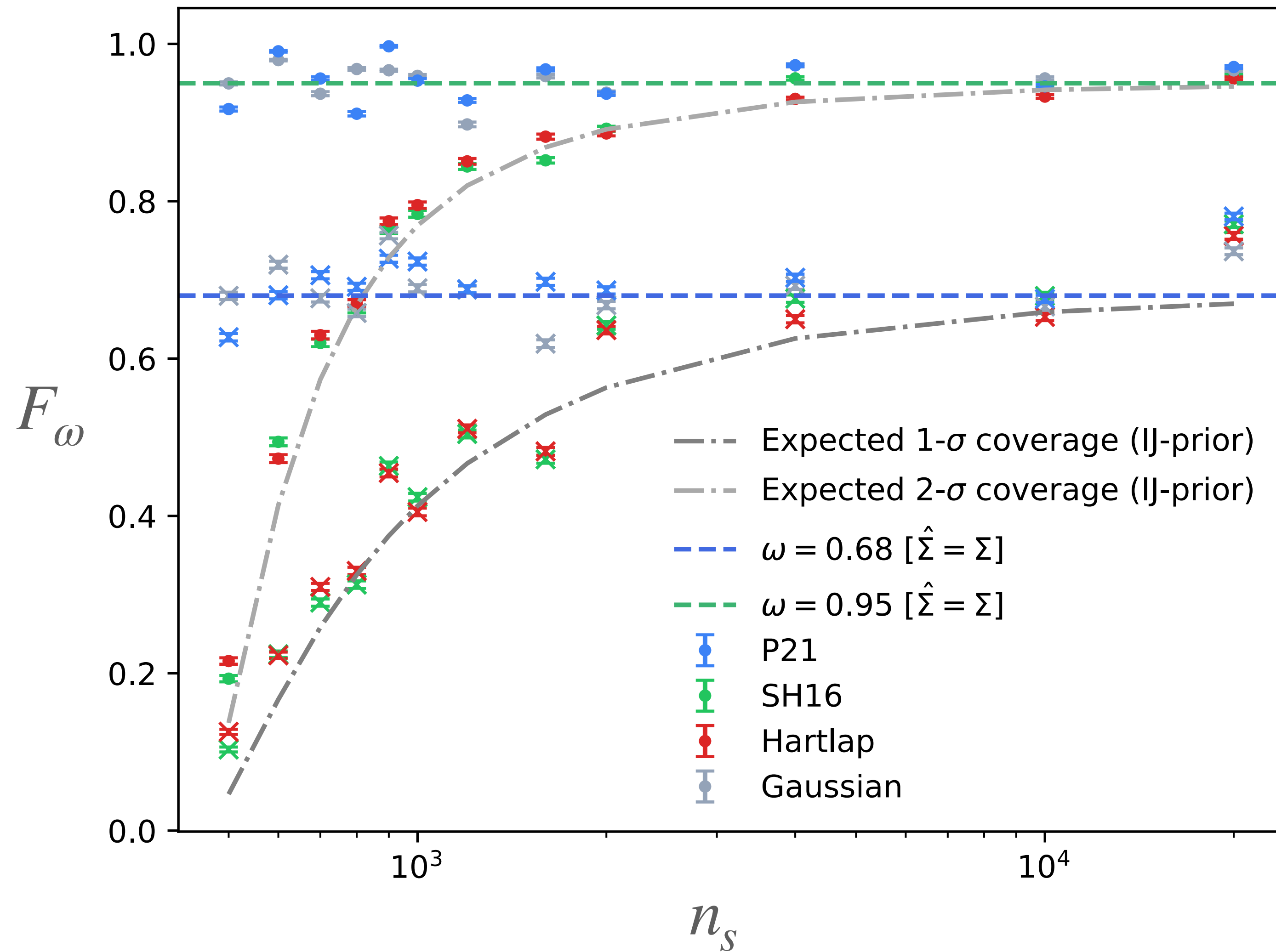
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Posterior widths - Gaussian likelihood analysis



Posterior coverages - Gaussian likelihood analysis



Analytic solution (Percival++21)

- Why did I only talk about Gaussian linear models?
 - First-order approximation to a non-linear model.
 - In high samples limit, approximately holds.

Analytic solution (Percival++21)

- Set a prior on the unknown covariance Σ then marginalise!

$$\begin{aligned} p(\pi | x, \hat{\Sigma}) &= \int d\Sigma \, p(\pi, \Sigma | x, \hat{\Sigma}) \\ &= \int d\Sigma \, p(x | \pi, \Sigma) p(\Sigma | \hat{\Sigma}) p(\pi) \end{aligned}$$

Analytic solution (Percival++21)

- Set a prior on the unknown covariance Σ then marginalise!
- Find parameterisation m of this prior $p(\Sigma) \propto |\Sigma|^{-\frac{m}{2}}$ by *matching* the scatter of the MLE with the expected width of the posterior

$$\langle \delta \hat{\pi} \delta \hat{\pi}^T \rangle_{x, \hat{\Sigma}} \approx \langle \delta \hat{\pi} \delta \hat{\pi}^T \rangle_{\pi|x, x}$$

- This gives the blue contours in our previous plots (see how this behaves!)

Summary

- The covariance Σ is a high-dimensional (n_x^2) object that is critical for $p(x | \pi)$
- There are many ways to estimate Σ , but even when we have accurate simulations, there is a catastrophic effect on our constraints (for realistic inference setups) due to the noisy estimate $\hat{\Sigma}$...
- This causes a scatter in the location and width of the parameter contours (the former being a worsened effect)
- We can account for the effect using the Dodelson-Schneider and Percival corrections, as well as a Frequentist-matching prior (Percival++21).
 - That said, these solutions are only accurate for Gaussian data x and linear models $x[\pi]$

Write down something you learned today and
something you need to hear more about next lecture

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